SESSION 4

Software Requirements Specification

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	Complete SRS Document (IEEE format)
Industry Mapping	Formal Documentation Standards
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Repository	[Add GitHub repository link]

Topics Covered in This Session

- Preparation of the Software Requirement Specification (SRS)
- IEEE SRS format: Introduction, Overall Description, Functional Requirements, Non-Functional Requirements
- Tools: MS Word / Google Docs / Overleaf

Revision History

Version	Date	Author	Description
0.1	[Date]	Team	Initial draft from Sessions 2 and 3
1.0	[Date]	Team	Complete SRS for review

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- 1. Introduction
- 2. Overall Description
- 3. Specific Requirements (3.1 External Interfaces, 3.2 Functional, 3.3 Non-Functional, 3.4 Data)
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1. Introduction

1.1 Purpose

This document specifies the requirements of the **AI Career Guidance & Skill Recommendation System**, version 1.0. It is intended for developers, testers, project guides and evaluators, and acts as the baseline for design, implementation and testing.

1.2 Scope

The system is a web application that lets students upload a resume, uses Claude AI to extract skills, education, projects and experience, compares the profile with predefined career requirements, and recommends suitable roles. It also provides skill gap analysis, a personalised learning roadmap, course recommendations, an analytics dashboard and an AI career chatbot. Admins manage the career and course data. The system does not provide job listings, payments or mobile apps in this version.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
API	Application Programming Interface
JWT	JSON Web Token used for authentication
MERN	MongoDB, Express, React, Node.js
Skill gap	Skills required for a career that the student does not yet have
Match score	Weighted percentage showing how well a profile fits a career
Roadmap	Ordered learning plan to close skill gaps
Claude	Large language model accessed through the Claude API

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- Session 2: Problem Statement Document.
- Session 3: Requirement Gathering Report.
- Claude API documentation (Anthropic); MongoDB and Express documentation.

1.5 Overview

Section 2 describes the product context, users and constraints. Section 3 lists specific functional, non-functional, interface and data requirements. Section 4 holds supporting appendices.

2. Overall Description

2.1 Product Perspective

The product is a new, self-contained web system with a three-tier structure: React client, Node.js/Express server, and MongoDB database. It depends on two external services: the Claude API for language understanding and generation, and a Python + Pandas analytics module for dashboard statistics.

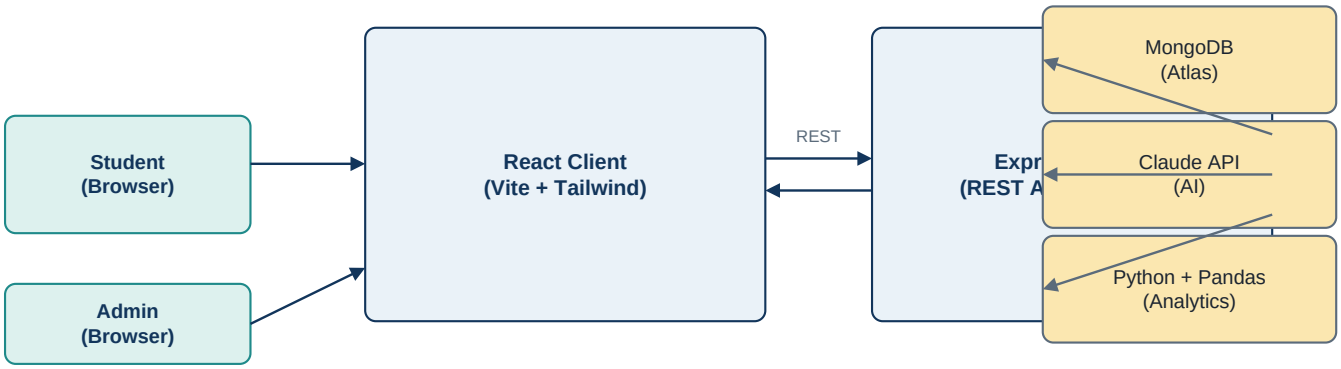

Figure 1: System context

2.2 Product Functions (summary)

- Register, login and manage profile.
- Upload resume and obtain AI-extracted profile data.
- Recommend careers with match scores.
- Analyse skill gaps for a selected career.
- Generate a personalised roadmap and recommend courses.
- Show analytics: in-demand skills, popular careers, common gaps.
- Answer career questions through an AI chatbot.
- Allow admins to manage careers, skills, courses and users.

2.3 User Classes and Characteristics

User class	Characteristics	Technical skill
Student	Main user; uploads resume, views guidance, uses chatbot	Basic web use
Admin	Maintains careers, skills, courses; views analytics	Moderate
Placement officer (view only)	Reads dashboard analytics	Basic

2.4 Operating Environment

- Client: modern browsers (Chrome, Edge, Firefox, Safari), desktop and mobile screens.
- Server: Node.js 18+ with Express; MongoDB 6+; Python 3.10+ for analytics.
- Hosting: any cloud platform supporting Node.js and MongoDB Atlas.

2.5 Design and Implementation Constraints

- Technology stack fixed as React, Node.js/Express, MongoDB, Claude API, Python/Pandas.
- API keys must be stored only on the server in environment variables.
- Resume files limited to PDF/DOCX, maximum 5 MB.
- Must comply with data protection practice: consent, secure storage, deletion on request.
- Development follows the Git workflow of Session 1.

2.6 Assumptions and Dependencies

- Users have internet access and a resume in digital form.
- Claude API is available and within usage limits.
- Career requirement data is entered and kept current by admins.
- AI outputs are guidance, not a guarantee of employment.

3. Specific Requirements

3.1 External Interface Requirements

Interface	Description
User interface	Responsive web UI with pages: Login/Register, Dashboard, Resume Upload, Career Recommendations, Skill Gap, Roadmap, Courses, Analytics, Chatbot, Admin Panel. Consistent navigation, clear error messages and loading indicators during AI processing.
Hardware interface	No special hardware. Standard device with a browser; file upload needs local storage access.
Software interfaces	Claude API (HTTPS, JSON); MongoDB via Mongoose; Python analytics via scheduled script or internal service; PDF/DOCX text extraction libraries.
Communication interfaces	HTTPS with REST/JSON between client and server; JWT in Authorization header; CORS restricted to the client origin.

3.2 Functional Requirements

FR-A: Authentication and Profile

ID	Requirement	Priority
FR-01	The system shall let a user register with name, email and password.	Must
FR-02	The system shall authenticate users and issue a JWT on successful login.	Must
FR-03	The system shall store passwords only as salted hashes.	Must
FR-04	The system shall restrict admin functions to users with the admin role.	Must
FR-05	The system shall let a student edit profile details, interests and target career.	Should

FR-B: Resume Analysis

ID	Requirement	Priority
FR-06	The system shall accept resume upload in PDF or DOCX format up to 5 MB and reject other files.	Must
FR-07	The system shall extract text from the uploaded file.	Must
FR-08	The system shall use Claude to extract skills, education, projects and experience as structured data.	Must
FR-09	The system shall display extracted data and allow the student to correct it.	Should
FR-10	The system shall keep previous resume versions and let the student delete them.	Should

FR-C: Career Recommendation

ID	Requirement	Priority
FR-11	The system shall compare the student profile with stored career requirements.	Must
FR-12	The system shall compute a match score per career = sum of importance weights of matched skills divided by sum of all weights for that career, as a percentage.	Must
FR-13	The system shall list at least the top 3 careers ranked by match score with a short AI-written explanation.	Must
FR-14	The system shall allow the student to choose any career as the target.	Should

FR-D: Skill Gap, Roadmap and Courses

ID	Requirement	Priority
FR-15	The system shall list matched skills and missing skills for the selected career.	Must
FR-16	The system shall classify each missing skill by priority (high, medium, low) using its importance weight.	Should
FR-17	The system shall generate a roadmap with ordered steps, estimated duration and a goal for each step using Claude.	Must
FR-18	The system shall let the student mark roadmap steps as complete and show overall progress.	Should
FR-19	The system shall recommend at least one course or resource for every missing skill.	Should

ID	Requirement	Priority
FR-20	The system shall allow roadmap export as PDF.	Could

FR-E: Analytics Dashboard

ID	Requirement	Priority
FR-21	The system shall show the most demanded skills across all careers.	Should
FR-22	The system shall show popular career choices among users.	Should
FR-23	The system shall show the most common skill gaps among users.	Should
FR-24	The system shall show the skill requirements for each career.	Should
FR-25	Analytics shall use aggregated, non-identifiable data only.	Must

FR-F: AI Career Chatbot

ID	Requirement	Priority
FR-26	The system shall provide a chat interface for career, skill, roadmap and preparation questions.	Should
FR-27	The chatbot shall use the student profile and recent chat history as context.	Should
FR-28	The system shall store chat history per student and allow clearing it.	Could
FR-29	The chatbot shall decline questions unrelated to careers and learning.	Should

FR-G: Administration

ID	Requirement	Priority
FR-30	Admin shall add, edit and delete careers with required skills, weights and levels.	Must
FR-31	Admin shall add, edit and delete courses linked to skills.	Should
FR-32	Admin shall view and deactivate user accounts.	Should

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Pages load within 3 s on a normal broadband connection; resume analysis completes within 15 s; chatbot replies within 8 s.
NFR-02	Scalability	Support at least 100 concurrent users; stateless server design allows horizontal scaling.
NFR-03	Security	HTTPS for all traffic; bcrypt password hashing; JWT expiry; input validation and sanitisation; rate limiting on AI endpoints; no API keys in client code.
NFR-04	Privacy	Resume data visible only to its owner and system; consent at upload; data deleted on request within 24 hours.
NFR-05	Reliability	Graceful handling of Claude API errors with retry and clear user message; no data loss on failure.
NFR-06	Availability	Target 99% uptime during working periods.
NFR-07	Usability	A first-time student completes upload to recommendation in under 5 minutes without help; responsive layout; accessible colour contrast.
NFR-08	Maintainability	Modular code, ESLint rules, README and API documentation; separate service layer for AI calls.
NFR-09	Portability	Runs on Windows, Linux and macOS development machines; containerisation optional.
NFR-10	Accuracy	Skill extraction accuracy of at least 85% on a sample of test resumes; AI output grounded in stored career data.

3.4 Data Requirements

Collection	Key fields
users	name, email, passwordHash, role, targetCareer, createdAt
resumes	userId, fileName, rawText, parsed {skills, education, projects, experience}, uploadedAt
skills	name, category, aliases
careers	title, description, requiredSkills [{skillId, weight, minLevel}]
recommendations	userId, resumeId, results [{careerId, matchScore, reason}], createdAt
skillGaps	userId, careerId, matched[], missing[], priority
roadmaps	userId, careerId, steps [{title, skillId, durationWeeks, status}]
courses	title, provider, url, skillId, level, cost
chatSessions	userId, messages [{role, text, time}]

4. Appendices

Appendix A: Requirement Traceability

User story (Session 3)	Functional requirements	Use case (Session 5)
US-01	FR-01 to FR-04	Register / Login
US-02, US-03	FR-06 to FR-10	Upload Resume, Analyze Resume
US-04	FR-11 to FR-14	View Career Recommendations
US-05	FR-15, FR-16	View Skill Gap Analysis
US-06, US-08	FR-17, FR-18	View Personalized Roadmap
US-07	FR-19	View Course Recommendations
US-09	FR-26 to FR-29	Chat with Career Bot
US-12	FR-21 to FR-25	View Analytics Dashboard
US-13, US-14	FR-30 to FR-32	Manage Careers, Courses, Users

Appendix B: Match Score Example

Career: Full Stack Developer with weights HTML/CSS 2, JavaScript 3, React 3, Node.js 3, MongoDB 2, Git 1 (total 14). A student who has JavaScript, React, Git and HTML/CSS scores (3 + 3 + 1 + 2) / 14 = 64%. Missing skills: Node.js and MongoDB, so the roadmap starts with Node.js (high priority).